

A BAP SUBCASE OF POINT SEPARATION FOR FREE UPPER- p -ESTIMATE BANACH LATTICES

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STATUS

This is a literature-implied partial answer to Remark 6.2 of Jardon-Sanchez–Laustsen–Taylor–Tradacete–Troitsky [1]. The source asks whether real-valued lattice homomorphisms on $\text{FBL}^{\mathcal{D}}[E]$ separate points, with the free upper p -estimate lattice singled out as already unclear. Combining the later finite-dimensional representation and approximation machinery of Oikhberg [2] gives a positive answer for $\text{FBL}^{\uparrow p}[E]$ whenever E has the bounded approximation property. The unrestricted case remains open.

SOURCE QUESTION

In Remark 6.2, after giving the explicit norm formula for the free p -convex Banach lattice, the source says:

$$= \|Rf\|_{L_p(h\,d\mu)} \leq M^{(p)}(X) \|T\| \|f\|_p. \quad \square$$

Remark 6.2. We do not know of an explicit formula for the norm of the free Banach lattice with upper p -estimates, even if the constant is one. In fact, we do not know whether this space — or $\text{FBL}^{\mathcal{D}}[E]$ in general — can be realized as a

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lattice of functions on the dual ball of E^* . Formulating the latter question more rigorously, we ask: Is it true that lattice homomorphisms from $\text{FBL}^{\mathcal{D}}[E]$ to $\mathbb{R}$ separate the points of $\text{FBL}^{\mathcal{D}}[E]$?

Thus the concrete subquestion addressed here is whether the canonical real-valued lattice homomorphisms

$$\widehat{x^*}: \text{FBL}^{\uparrow p}[E] \rightarrow \mathbb{R}, \quad x^* \in E^*,$$

separate points.

SUPPORTING INGREDIENTS FROM OIKHBERG

Oikhberg [2] studies the free upper p -estimate lattice $\text{FBL}^{\uparrow p}[E]$, $1 < p < \infty$. We use two consequences from that paper.

First, for finite-dimensional F , the formal identities

$$\text{FBL}[F] \longrightarrow \text{FBL}^{\uparrow p}[F] \longrightarrow \text{FBL}^{(\infty)}[F]$$

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are contractive, and $\text{FBL}^{\uparrow p}[F]$ is represented as a lattice of continuous positively homogeneous functions on the sphere of F^* ; see the discussion around Remark 2.5 and Corollary 2.7 of [2].

Second, the proof of Theorem 2.9 of [2], in the same manner as Theorem 2.8 there, uses finite-rank approximants $u_\alpha : E \rightarrow E_\alpha$ and the induced lattice homomorphisms

$$\overline{u_\alpha} : \text{FBL}^{\uparrow p}[E] \rightarrow \text{FBL}^{\uparrow p}[E_\alpha], \quad \overline{j_\alpha} : \text{FBL}^{\uparrow p}[E_\alpha] \rightarrow \text{FBL}^{\uparrow p}[E],$$

where $j_\alpha : E_\alpha \hookrightarrow E$, and $\overline{j_\alpha} \overline{u_\alpha} \rightarrow I$ point-norm when (u_α) is a bounded approximation of I_E .

IDEA OF THE PROOF

The source obstruction is that a completion of the algebraic free lattice could, a priori, contain a “ghost” element invisible to every point evaluation on E^* . Bounded approximation rules this out by testing the ghost through finite-dimensional shadows. In finite dimensions the later functional representation says point evaluations are faithful. If an element is killed by every $\widehat{x^*}$ on the large free lattice, then every finite-dimensional shadow $\overline{u_\alpha} z$ is killed by all point evaluations on $\text{FBL}^{\uparrow p}[E_\alpha]$, hence is zero. The finite-rank shadows approximate the identity, so $z = 0$.

PARTIAL THEOREM

Theorem 1. *Let $1 < p < \infty$, and let E be a Banach space with the bounded approximation property. Then the real-valued lattice homomorphisms*

$$\{\widehat{x^*} : \text{FBL}^{\uparrow p}[E] \rightarrow \mathbb{R} \mid x^* \in E^*\}$$

separate the points of $\text{FBL}^{\uparrow p}[E]$. Equivalently, the canonical map from $\text{FBL}^{\uparrow p}[E]$ into the space of continuous positively homogeneous functions on the dual ball is injective.

Proof. We first record the finite-dimensional case. Let F be finite-dimensional and let L_F be the algebraic free vector lattice generated by the functions δ_x , $x \in F$, on F^* . The construction of $\text{FBL}^{\uparrow p}[F]$ gives

$$\|f\|_\infty \leq \|f\|_{\uparrow p} \quad (f \in L_F).$$

The identity map on F induces a contractive lattice homomorphism $\text{FBL}[F] \rightarrow \text{FBL}^{\uparrow p}[F]$, so

$$\|f\|_{\uparrow p} \leq \|f\|_{\text{FBL}[F]}.$$

Since F is finite-dimensional, the usual functional representation of $\text{FBL}[F]$ gives $\|f\|_{\text{FBL}[F]} \leq C_F \|f\|_\infty$ on L_F ; for instance one may take $C_F = \dim F$ from the standard $\pi_1(I_F) \leq \dim F$ estimate. Hence the $\uparrow p$ -norm and the supremum norm are equivalent on L_F . The completion $\text{FBL}^{\uparrow p}[F]$ therefore embeds injectively into the corresponding supremum-norm completion. Thus the point evaluations, equivalently the homomorphisms $\widehat{y^*}$ for $y^* \in F^*$, separate points of $\text{FBL}^{\uparrow p}[F]$.

Now let E have the λ -bounded approximation property. Choose a net of finite-rank operators $u_\alpha : E \rightarrow E$ such that $\sup_\alpha \|u_\alpha\| \leq \lambda$ and $u_\alpha x \rightarrow x$ for every $x \in E$. Put $E_\alpha = u_\alpha(E)$, and write $u_\alpha : E \rightarrow E_\alpha$ for the same map and $j_\alpha : E_\alpha \hookrightarrow E$ for inclusion. By the universal property of $\text{FBL}^{\uparrow p}$, these operators induce lattice homomorphisms

$$\overline{u_\alpha} : \text{FBL}^{\uparrow p}[E] \rightarrow \text{FBL}^{\uparrow p}[E_\alpha], \quad \overline{j_\alpha} : \text{FBL}^{\uparrow p}[E_\alpha] \rightarrow \text{FBL}^{\uparrow p}[E].$$

The operators $A_\alpha = \overline{j_\alpha} \overline{u_\alpha}$ have uniformly bounded norms. On the dense algebraic free vector lattice $\text{FVL}[E]$, A_α sends a lattice expression in $\delta_{x_1}, \dots, \delta_{x_n}$ to the same expression in $\delta_{u_\alpha x_1}, \dots, \delta_{u_\alpha x_n}$, which converges in norm by continuity of lattice operations. Therefore $A_\alpha \rightarrow I$ point-norm on all of $\text{FBL}^{\uparrow p}[E]$.

Let $z \in \text{FBL}^{\uparrow p}[E]$ satisfy $\widehat{x^*}(z) = 0$ for every $x^* \in E^*$. Fix α and $y^* \in E_\alpha^*$. Then

$$\widehat{y^*}(\overline{u_\alpha} z) = \widehat{u_\alpha^* y^*}(z) = 0.$$

The finite-dimensional case applied to E_α gives $\overline{u_\alpha} z = 0$. Hence $A_\alpha z = 0$ for every α . Since $A_\alpha z \rightarrow z$, we conclude that $z = 0$. This proves point separation. $\square$

SCOPE AND LIMITATIONS

This does not settle the general question from [1]. It covers the free upper p -estimate lattice only under the bounded approximation property assumption on E . It also does not address arbitrary convexity conditions $\mathcal{D}$. The result is best viewed as a BAP/localization corollary of [2], not as an independent full resolution.

VERIFICATION NOTES

Cheap run indexes were searched for arXiv 2101.03510, the title, free Banach lattices, upper p -estimates, point separation, and lattice homomorphisms. Relevant later local sources 2210.00614, 2303.05209, 2404.06497, 2601.11056, and 2605.28988 were checked. The exact BAP point-separation corollary was not found as an existing run packet.

REFERENCES

- [1] H. Jardon-Sanchez, N. J. Laustsen, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices under convexity conditions*, Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Mat. RACSAM 116 (2022), Paper No. 15; arXiv:2101.03510.
- [2] T. Oikhberg, *Geometry of unit balls of free Banach lattices, and its applications*, arXiv:2303.05209.